SQL Injection Scanner

Syntax Hub Cybersecurity Internship — Task 3

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Technology: Python
Platform: Google Colab

1. Introduction

The SQL Injection Scanner is a Python-based cybersecurity project that identifies common SQL error indicators in text responses.

2. Objectives

- Learn the basics of SQL Injection detection.
- Identify common SQL error messages.
- Analyze sample responses using Python.
- Generate JSON and CSV reports.

3. Tools Used

- Python
- Google Colab
- Regular Expressions
- JSON and CSV modules

4. Methodology

The program checks simulated response text for predefined SQL error patterns. It records the results and saves them in JSON and CSV files.

5. Testing and Results

Four simulated test cases were used: a normal response, a simulated SQL syntax error, a simulated Oracle database error, and a generic error message.

The program behaved as expected for all four cases in the offline demonstration.

6. Limitations

This prototype analyzes simulated responses only. It does not send HTTP requests or scan real websites, and its results do not confirm a real vulnerability.

7. Conclusion

This project provided practical experience with Python, regular expressions, basic security testing concepts, and generating structured reports.

8. Future Improvements

- Testing in an authorized lab
- Controlled HTTP request handling
- Rate limiting and concurrency
- Improved vulnerability reporting